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15 DECEMBER 1961
SUPERSEDING
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1 MARCH 1956

MILITARY SPECIFICATION

DIRECTION FINDER GROUP AN/APA-69A AND AN/APA-69C (U)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope—The equipment specified herein has been designed for direction finding in the frequency range of 180 to 10750 mc for the AN/APA-69A equipments and 50 to 10750 mc for the AN/APA-69C equipments. The AN/APA-69 series shall consist of 3 interchangeable rotating antennas, an indicator and a console control unit for use with Countermeasures Receiving Set AN/ALR-8 or similar receivers on Naval or other Military aircraft to permit true bearings to be taken on sources of radar and radio signals. The equipment shall display each receiver signal as an antenna pattern on a CRT with distance from the center of the CRT screen representing magnitude of the signal, and position around the CRT representing true bearing on the signal source.

1.2 Classification—The Direction Finder Group, Radar, AN/APA-69 series shall consist of the following units.

AN/APA-69A COMPOSITION

Item	Type Designation	Applicable Paragraphs
Antenna Assembly	AS-435/APA-69	3.4.8
Antenna Assembly	AS-434/APA-69	3.4.9
Antenna Assembly	AS-436/APA-69	3.4.10
Antenna Reflector	AT-182/APA-69	3.4.11
Antenna Drive	TG-8A/APA-69	3.4.1
Antenna Coupler	CU-174/APA-69	3.4.12
Azimuth-Panoramic Indicator	IP-81A/APA-69	3.4.3
Direction Finder Control	C-527A/APA-69	3.4.4
Mounting	MT-1005/APA-69	3.4.13

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AN/APA-69C COMPOSITION

Item	Type Designation	Applicable Paragraphs
Antenna Assembly	AS-775/APA-69	3.4.6
Antenna Assembly	AS-676/APA-69	3.4.7
Antenna Assembly	AS-436/APA-69	3.4.10
Antenna Reflector	AT-182/APA-69	3.4.11
Antenna Drive	TG-50/APA-69	3.4.2
Antenna Coupler	CU-174/APA-69	3.4.12
Azimuth-Panoramic Indicator	IP-81A/APA-69	3.4.3
Direction-Finder Control	C-1321/APA-69	3.4.5
Mounting	MT-1005/APA-69	3.4.13

FSC 5826

2. APPLICABLE DOCUMENTS

2.1 General—The following documents of the issue in effect on the date of invitations for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting; and Mounts Vibration (For Use with Electronic Equipment in Aircraft)
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-5400	Electronic Equipment, Aircraft, General Specifications for
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-I-6181	Interference Limits and Tests; Aircraft Electrical and Electronic Equipment
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-T-18303	Test Procedures; Preproduction and Inspection for Aircraft Electronic Equipment, Format for
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-E-4682	Electron Tubes, Choice and Application of

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Bureau of Naval Weapons

EI-207	Control Panels, Electrical and Electronic Equipment, Rack or Console Mounted, Installation of

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single-Phase AC, or 115/200-Volt 3-Phase AC, or Combinations
SAR-300	Change Proposals, Material Changes and Electronic Material Bulletins, Preparation of

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials—In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hr should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval—Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices—Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction—

3.2.1 General—This equipment shall fulfill all applicable design and performance requirements of Specification MIL-E-5400—Class 1 unless otherwise specified herein.

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3.2.2 Total Weight—The total weight of the equipment, excluding cable, shall be a minimum consistent with good design and shall not exceed 86 lb. for the AN/APA-69A equipments or 84 lb for the AN/APA-69C equipments. Total weight shall include all major units and accessories (less cable) necessary to fulfill the scope of the equipment as stated in paragraph 1.1.

3.2.3 Operating Life—

3.2.3.1 Operational Stability—The equipment shall operate with optimum performance for 25 hr, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.3.2 Total Operating Life—The equipment shall have a minimum total operating life of 2,000 hrs with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturers.

3.2.4 Cabling and Connections—

3.2.4.1 Cables and Connectors—The equipment shall be designed for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and that portion of the connectors which is to be attached to the cables shall not be supplied as part of the equipment.

3.2.4.2 Interconnection Cabling—This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements and provided assembly of the cable to its plugs may be easily accomplished.

3.2.4.3 Input Power Connection—Input power connections shall be in accordance with MS-25245.

3.2.5 Control Panels—All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.6 Interchangeability—The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.7 Interference Control—The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.8 Maintenance Provisions and Field Testing—Provision for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.9 Nomenclature and Nameplates—Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.10 Standard Conditions—The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

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Temperature	Room ambient (30° ± 10° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input power voltage	115 ± 1.0 vac and/or 27.5 ± 0.5 vdc

3.2.11 Service Conditions—The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations as specified in Specification MIL-E-5400.

3.2.12 Primary Input Power Requirements—The AN/APA-69 series shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified herein. The power required shall not exceed:

(1) AN/APA-69A equipment

A-C Power (Single-Phase) 115 v, Category "B" 38 va
D-C Power 28-v Category "B" 3.5 amp
Power Required from AN/ALR-8 Power Supply—6 ma at 150 vdc
44 ma at 300 vdc; 1.6 amp at 6.3 vac

(2) AN/APA-69C equipment

A-C Power (Single-Phase) 115 v, Category "B" 49 va
D-C Power 28-v Category "B" 5.5 amp
Power Required from AN/ALR-8 Power Supply—6 ma at 150 vdc
44 ma at 300 vdc; 1.6 amp at 6.3 vac

3.3 Performance—The equipment shall meet the performance requirements as specified below:

3.3.1 Indication—The equipment shall present a visual display on the screen of the cathode ray tube. The radial deflection of the electron beam, shall be proportional to the signal strength received, and the beam shall rotate in synchronism with the antenna producing a plot of the radiation pattern of the antenna.

3.3.2 Accuracy of Indication—The accuracy of indication shall be divided into two parts. One covering instrument error and the other pattern error defined as follows:

(1) Instrument Error of Indication—This error shall be defined as the error between the actual angular position of the antenna drive shaft, and the indicated position on the indicator. This error shall not exceed ±2.5 degrees for any angular position of the antenna drive shaft.

(2) Antenna Pattern Error—The accuracy of angular position of the major lobe of each of the AN/APA-69 antennas in an azimuth plane perpendicular to the antenna drive shaft at the antenna and with respect to the mechanical zero index shall be as follows:

(a) AS-775/APA-69—The accuracy of the major lobe with horizontal polarization shall be within ±10 degrees throughout the frequency range from 50 to 320 mc. Pattern checks shall be made at 60, 130, 200, and 310 mc.

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(b) AS-676/APA-69—With this combination of antennas the azimuth accuracy of the major lobe of the low frequency horizontally polarized antenna shall be within ±7 degrees throughout the frequency range between 180 and 1500 mc. Pattern checks shall be made at 200, 500, 800, 1100, and 1400 mc and shall be accurate to within ±5 degrees. The major lobe of the low frequency vertically polarized antenna shall have similar accuracy throughout the same frequency range and meet the same accuracy requirements specified for the horizontally polarized low frequency antenna. The major lobe of the medium frequency antenna, either vertically or horizontally polarized, shall be accurate to within ±5 degrees throughout the frequency range between 1000 and 5000 mc. Pattern checks shall be made at 1500, 2500, 3500, and 4500 mc and shall be accurate to within ±3 degrees.

(c) AS-435/APA-69—Accuracy of the major lobe of the antenna shall be within ±7 degrees at all points throughout the frequency spectrum of 250 to 1000 mc for either vertical or horizontal polarization. Pattern checks made at 300, 400, 800, and 1000 mc shall be accurate within ±5 degrees.

(d) AS-434/APA-69—Accuracy of the major lobe of the antenna shall be within ±4 degrees throughout the frequency range from 1000 to 5000 mc for either vertical or horizontal polarization. Pattern checks made at 1000, 2200, 3300, and 4200 mc shall be accurate within ±3 degrees.

(e) AS-436/APA-69—Antenna with AT-182/APA-69 Reflector—With this combination, the azimuth accuracy of the major lobe of the antenna shall be within ±5 degrees throughout the range from 4300 to 10300 mc for either vertical or horizontal polarized signals. Measurements taken at 4500, 6600, 9200, and 10350 mc shall be accurate within ±3 degrees. To maintain this accuracy, a mechanical shift in the position of the AT-182/APA-69 reflector or its mounting flange is permissible to correct for an approximately 4-degree constant displacement of the beam which appears in the AS-436/APA-69 and the AT-182/APA-69 combination.

3.3.3 Accuracy Measurement—This accuracy of bearing shall persist whether the radar signal is vertically or horizontally polarized, assuming pure polarization of received signal. The accuracy of bearing indication for the AS-775/APA-69 antenna shall be ±10 degrees, and for the AS-676/APA-69 antenna, the accuracy shall be within ±5 degrees over 80 percent of the frequency band with either polarization. Bearing errors for the AS-434/APA-69, AS-435/APA-69 and AS-436/APA-69 antennas shall not exceed ±6 degrees.

3.3.4 Antenna Rotation Speed—The antennas shall be satisfactorily balanced and mechanically rigid to rotate, without excessive vibration or damage, at speeds up to 300 rpm at ambient temperature of 20° C. This speed shall be made variable by a speed control on the control unit. It shall be satisfactory to measure the average speed after 5 min at a temperature of –55°C. The average speed should be at least 150 rpm.

3.3.5 Antenna Gain—

AS-775/APA-69—Not down more than 25 ± 3 db from that of a lossless half-wave matched dipole located one-quarter wavelength in front of a large ground plane throughout the frequency range from 50 to 320 mc.

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AS-676/APA-69—The low frequency coverage shall not be more than 7.5 db below a single dipole in free space at any point between 180 and 500 mc, and not down more than 5.0 db from that of a single dipole in free space at any point between 500 and 1500 mc. The high frequency coverage shall not be more than 5.0 db below a single dipole in free space at any point within the frequency range from 1000 to 5000 mc.

AS-435/APA-69—Not more than 5 db down from that of a dipole in free space at any point within the frequency range from 140 to 1000 mc.

AS-434/APA-69—No less than a dipole in free space at any point within the frequency range from 1000 to 5000 mc.

AS-436/APA-69—At least 5 db better than a dipole in free space at every point throughout the frequency range from 5000 to 10750 mc.

3.3.6 Antenna Characteristics—

AS-775/APA-69—At no point in the frequency range between 65 and 300 mc shall the standing wave ratio exceed 2.5:1 for horizontal polarization.

AS-676/APA-69—The impedance characteristics of the antennas measured at the output of the rotary point, with no interconnecting cables between the output fitting and the impedance measuring equipment, shall meet the following conditions when operated into a 50-ohm line:

Low Frequency Antennas—Within the frequency range between 180 and 1500 mc, the following ratios shall be considered. The portion of the area under the curve (VSWR vs FREQUENCY) which is above the 5:1 line and the portion under the curve which is above the 10:1 line will be divided by the total area under the 5:1 line. For the horizontally polarized antenna, the 5:1 area ratio shall not exceed 0.30 and the 10:1 area ratio shall not exceed 0.05. For the vertically polarized antenna, the 5:1 area ratio shall not exceed 0.40 and the 10:1 area ratio shall not exceed 0.10.

Medium Frequency Antennas—Within the frequency range between 1000 and 5000 mc, the ratios as specified above shall not exceed for the horizontally polarized position of the feed, 0.90 for the 5:1 area ratio; 0.20 for the 10:1 area ratio. For the vertically polarized position of the feed, the ratios shall not exceed 0.90 for the 5:1 area ratio and 0.20 for the 10:1 area ratio.

AS-435/APA-69—At no point in the frequency range from 350 to 1000 mc shall the standing wave ratio exceed 8 to 1 for horizontal polarization. Production tests points shall be taken at 350, 500, and 1000 mc. The ratio of the portion of the area under and above the curve (standing wave ratio plotted against frequency) shall not exceed 0.29 for the horizontal antenna and 0.22 for the vertical antenna.

AS-434/APA-69—At no point in the frequency range from 1000 to 5000 mc shall the standing wave ratio of either the vertical or horizontal antenna exceed 10 to 1. Production test frequencies shall be taken at 1200, 3000, and 4500 mc; the voltage standing wave ratio should not exceed 6, 10, and 7, respectively, for vertical polarization and 8, 10, and 8, respectively, for horizontal polarization. The ratio of the portion above and below the curve shall not exceed 0.3 for either the horizontal or vertical antenna.

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AS-436/APA-69—At no point within the frequency range from 4450 to 10500 mc shall the standing wave ratio of this antenna exceed 4:1.

3.4 Detailed Requirements—

3.4.1 Antenna Drive TG-8A/APA-60A—The antenna drive unit shall be mounted on the aircraft for any one of the three AN/APA-69A antennas specified in paragraphs 1.2 and 3.3.2. The drive shall be rotatable by a motor mechanism for spinning the antenna in a horizontal plane at speeds specified in paragraph 3.3.4.

3.4.1.1 Form Factor and Mounting—The antenna drive unit shall be approximately 11.2-in. long, 8.2-in. wide, and 7.4-in. high. It shall mount on the aircraft by two flanges on each side of the unit and shall not be vibration mounted.

3.4.1.2 Weight—The weight of the antenna drive unit, excluding antennas and cables shall not exceed 14 lb.

3.4.1.3 Contents of the Unit—The antenna drive unit shall contain the following:

(1) Drive motor

(2) Driving gear system

(3) Sine, cosine generator

(4) Antenna mounting pedestal

(5) Speed variation switch

(6) Video polarization reversal relay

(7) Switch for disabling video polarization relay when using AS-436/APA-69 antenna

3.4.1.4 Electrical Connections—Electrical connections to the unit shall provide the following receptacles:

AN-3102A-18-8P—	Power and control cable
UG-290/U—	North, South video
UG-290/U—	East, West video
UG-290/U—	Video input

3.4.1.5 Controls—No controls shall be located on the unit except for a toggle switch for disabling the video phase reversal relay when the AS-436/APA-69 antenna is installed.

3.4.2 Antenna Drive TG-50/APA-69—The antenna drive unit shall be mounted on the aircraft for any one of the antennas specified in paragraph 3.3.2. The drive shall be rotatable by a motor mechanism for spinning the antenna in a horizontal plane at speeds specified in paragraph 3.3.4.

3.4.2.1 Form Factor and Mounting—The antenna drive unit shall be approximately 11.2-in. long, 8.2-in. wide, and 7.4-in. high. It shall mount on the aircraft by two flanges on each side of the unit and shall not be vibration mounted.

3.4.2.2 Weight—The weight of the antenna drive unit, excluding antennas and cables shall not exceed 14.9 lb.

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3.4.2.3 Contents of Unit—This unit shall contain the following:

(1) Drive motor

(2) Driving gear system

(3) Antenna mounting pedestal

(4) Video polarization reversal relay

(5) Switch for disabling video polarization relay when using AS-436/APA-69 or AS-775/APA-69 antenna assembly

(6) Sine, cosine generators

3.4.2.4 Electrical Connections—Electrical connections to the unit shall provide the following receptacles:

AN-3102A-18-8P—	Power and control cable
UG-290/U—	North, South video
UG-290/U—	East, West video
UG-290/U—	Video input

3.4.2.5 Controls—No controls shall be located on the unit except for a toggle switch for disabling the video phase reversal relay when the AS-436/APA-69 or the AS-775/APA-69 antenna is installed.

3.4.3 Azimuth—Panoramic Indicator IP-81A/APA-69—This unit shall meet the following requirements.

3.4.3.1 Function—The function of the azimuth indicator shall be a remote indicating device which shall receive the video and antenna position information from the antenna drive unit and present true bearing information of received signals on a cathode ray tube. This information shall be read directly in degrees on a graduated scale around the circumference of the face of the cathode ray tube. The scale shall be rotated by a servo system operating from the aircraft gyrocompass so that signal bearings are read in true bearing on the scale while aircraft heading can be read under a fixed index at the top of the scale. It is intended that this unit operate in conjunction with the AN/ALR-8 receiving equipments and shall provide either bearing information on received signals, or panoramic presentation of the signal spectrum obtained from the AN/ALR-8 receiving equipments. A switch shall be provided for switching the function of the indicator from direction-finding to panoramic. (The IP-81A/APA-69A is a replacement of a former type, i.e., IP-36/APA-69 amplifier power supply unit AM-256/APA-69 and ID-226/APR-9 or ID-226A/APR-9).

3.4.3.2 Form Factor and Mounting—This unit shall be constructed in accordance with Specification MIL-C-172 and shall be housed in a case not to exceed 10.3-in. wide, 7.7-in. high and 16.2-in. long. The overall length of the unit shall be approximately 19.2 in., including panel and plug projections. A detachable rubber light shield shall be provided and, with the light shield in place, the overall length of the unit shall be approximately 22.7 in. MT-1005/APA-69A mounting unit shall be supplied as part of this unit.

3.4.3.3 Contents—The unit shall contain the following assemblies, subassemblies and circuits:

(1) Three-inch CRT

(2) Horizontal and vertical deflection amplifiers

(3) Relays to permit switching the indicator function from panoramic to DF

(4) Intensity modulation amplifier

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(5) Pulse stretcher and baseline stabilizer for panoramic function

(6) Video amplifier and pulse stretcher for direction finding function

(7) Remote frequency indicating synchro and synchronizing circuits

(8) Alidade servo amplifier (driven from compass system)

(9) True bearing indicating alidade

3.4.3.4 Weight—The weight of this unit excluding its mounting base shall not exceed 28 lb.

3.4.3.5 Electrical Connections—Electrical connections shall be made to the rear of the case and shall consist of the following:

Receptacle	Function
Amphenol No. 97-3102A-36-405P or an approved equivalent	ALR-8 power and control input
AN-3102A-20-7P	APA-69 power and control input
AN-3102A-14S-2P	Panoramic input
UG-290/U	Video input
UG-290/U	Resolver video output
UG-290/U	Resolver E-W video input
UG-290/U	Resolver N-S video
AN-3102A-14S-5P	Compass servo input

3.4.3.6 Controls—The following controls shall be located on the front panel:

(1) Oscilloscope Focus Control

(2) Oscilloscope Intensity Control

(3) Alidade Lights Brilliance Control

(4) Frequency Indicator Reset Switch

(5) Bearing Indicating Scale, and Alidade

(6) Oscilloscope Trace Positioning Control

(7) Intensity Modulation Control

(8) Horizontal Gain Control for Panoramic Function

3.4.3.7 Panoramic Function:

3.4.3.7.1 Vertical Deflection Amplifier—The vertical deflection amplifier, without the pulse stretcher and base line stabilizer, shall have an amplification characteristic that is not more than 3 db below maximum response between 100 and 20000 cps as measured on the oscilloscope screen. Voltage amplification as measured in this manner shall be at least 30 db. Voltage amplification, with the pulse stretcher, and base line stabilizer circuits included, shall be such that the amplitude of a 10-usec, 500-prf negative pulse required to produce a 1-in. deflection on the screen of the CRT shall not be greater than 3-v peak. For the same test conditions, variation of the CRT vertical deflection across the horizontal sweep shall not exceed 2 db. When the pulse width at 500 prf is varied from 0.5 to 50 mc, the CRT deflection shall not vary more than 6 db. When the prf of a 10-usec pulse is varied from 50 to 2000, the CRT deflection shall not vary more than 4 db. The base line stabilizer shall maintain the position of the base line on oscilloscope screen substantially constant during intervals of high-level signal receptions. It

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shall also remove the effect of responses that produce deflections below the base line.

3.4.3.7.2 Horizontal Deflection Amplifier—The horizontal deflection amplifier shall be directly coupled as far as possible to minimize phase shift at the sweep frequency. The input R-C coupling network shall have a sufficiently large time constant to pass the low frequency horizontal sweep voltage from the mixer amplifier with a calculated phase shift of less than 1 degree. A control shall be provided on the front panel for setting the gain of the horizontal amplifier. With the control at the maximum position, the voltage gain of the amplifier as measured from the panoramic input receptacle to the deflection plates of the CRT shall be at least 23 db. The horizontal sweep deflection versus input voltage characteristic shall be linear to at least ±10 percent of the total deflection up to a deflection of 2.5 in.

3.4.3.7.3 Intensity Modulation Amplifier—A control shall be provided on the front panel to adjust the intensifier voltage. When the amplifier is conditioned for panoramic reception and with the control set at maximum the intensity modulation amplifier shall develop at least a 25-v pulse to the cathode of the CRT when a 1.5-v pulse is applied to the panoramic input receptacle pin A of J-1007. The output signal of the amplifier shall be limited to not more than 90 v.

3.4.3.8 Direction Finding Function

3.4.3.8.1 Video Amplifier—When a 10-usec, 500-prl positive pulse of 1 v is applied to the input of the video amplifier, the output of the resolver (in TG-8A/APA-69A or TG-50/APA-69C) shall be not less than 25 v. The high frequency response of the video amplifier shall be such that the output amplitude of 0.25-usec input pulse shall be not less than half the amplitude of a 10-usec pulse of the same input amplitude. The low frequency response shall be such that the trailing edge of a 50-usec input pulse shall not drop to less than half its maximum amplitude.

3.4.3.8.2 Deflection Amplifier—With no signal input, it shall be possible to center the beam on the CRT screen. The amplitude of the (positive or negative) pulse required at the input of the horizontal deflection amplifier (E-W video input) to produce a 1-in. deflection on the CRT shall not exceed 2.5 . With the same amplitude (positive or negative) pulse applied at the input of the vertical deflection amplifier (N-S video input) that is required to produce a 1-in. deflection on the horizontal amplifier, it shall be possible to set the vertical deflection to 1-in. by adjusting the circularity control.

3.4.3.8.3 Deflection Amplifier Characteristics—When the intensity modulation amplifier is conditioned for DF reception, the amplifier shall receive its signal from the output of the DF video amplifier. When in this position, with a 1.5-v pulse of 500 prf fed to the input of the video amplifier, J-1008, the output pulse to the cathode of the CRT shall develop at least 25 v when the intensity amplifier gain control is set at maximum. As the amplitude of the input pulse is increased, the output pulse of the intensity amplifier shall be limited to not more than 60 v.

3.4.3.9 True Bearing Torque and Amplifier Assemblies—The true bearing torque assembly shall operate from information supplied from a gyro fluxgate compass rotating the azimuth scale on the indicator such that true bearing is always indicated on the scale with the heading of the aircraft shown by reading the compass driven scale opposite a fixed index mark at the top of the scale. The torque amplifier shall provide the necessary amplification to the compass position voltages to drive the autosyn scale positioning torque unit. The alidade, driven by the servo system shall provide true heading of the aircraft to ±1.5 degrees assuming no error in the aircraft compass system.

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3.4.3.10 Remote Frequency Indicating Synchro and Synchronizing Circuits—The synchronizing light shall illuminate with operation of the limit switch at the low frequency end of any AN/ALR-8 tuner. This shall permit alignment of the remote frequency counter dial, at the indicator unit, to the indicated frequency of an r-f tuner to an accuracy of at least ±5 mc. This accuracy shall be maintained throughout the tuning range of any AN/ALR-8 r-f tuners.

3.4.3.11 Indicators—The following indicators shall be provided on the front panel:

(1) Oscilloscope screen for panoramic or DP presentation of a received signal.

(2) Remote frequency counter dial

(3) Synchronizing light (for remote frequency dial alignment)

(4) True bearing alidade

3.4.4 Direction Finder Control C-527A/APA-69—This unit shall meet the following requirements:

3.4.4.1 Function—The control unit shall provide all controls necessary for operation of the AN/APA-69A equipment when using the IP-81A/APA-69A azimuth-panoramic indicator with the AN/ALR-8 receiver.

3.4.4.2 Form Factor and Mounting—The unit shall conform to the requirements of Specification MIL-C-6781. The length of the panel shall not exceed 4.5 in. and the depth of the cover box shall not exceed 6.4 in. The unit shall mount in accordance with Specification EI-207. Fasteners shall be in accordance with Specification MIL-F-25173.

3.4.4.3 Weight—The weight of this unit excluding cables shall not exceed 5 lb.

3.4.4.4 Controls—The unit shall be marked and contain the following controls:

(1) "DF-PAN" two position selector switch

(2) "PWR-OFF" power switch

(3) "ANT-SPEED" control rheostat

(4) "HOR-VERT" two position antenna selector switch

(5) "GAIN" direction finder control

3.4.4.5 Electrical Connections—Electrical connections shall be made to the rear of the unit in accordance with Specification MIL-C-6781 and shall consist of one AN-3102A-24-7P receptacle.

3.4.5 Direction Finder Control C-1321/APA-69—This unit shall meet the following requirements:

3.4.5.1 Function—The control unit shall provide all controls necessary for operation of the AN/APA-69C equipment when using the IP-81A/APA-69C azimuth-panoramic indicator with the AN/ALR-8 receiver.

3.4.5.2 Form Factor and Mounting—The unit shall conform to the requirements of Specification MIL-C-6781. The length of the panel shall not exceed 4.5 in. and the depth of the cover box shall not exceed 6.4 in. The unit shall mount in accordance with Specification EI-207. Fasteners shall be in accordance with Specification MIL-F-25173.

3.4.5.3 Weight—The weight of this unit excluding cables shall not exceed 4 lb.

3.4.5.4 Controls—The unit shall be marked and contain the following controls:

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(1) "DF-PAN" two position selector switch

(2) "PWR-OFF" two position power switch

(3) "ANT-SPEED" "ZERO" control rheostat

(4) POLAR-LH-LV-MH-MV-V-H- six position rotary antenna selector switch

(5) "GAIN" direction finder control

3.4.5.5 Electrical Connections—Electrical connections shall be made to the rear of the unit in accordance with Specification MIL-C-6781 and shall consist of one AN-3102A-24-7P receptacle.

3.4.6 Antenna Assembly AS-775/APA-69—This unit shall meet the following requirements in addition to the applicable requirements of paragraph 3.3.

3.4.6.1 Function—This assembly shall mount on the TG-50/APA-69 drive unit and together with the antenna position information supplied from the drive unit, shall furnish bearing information on horizontally polarized signals within the frequency range from 50 to 320 mc.

3.4.6.2 Form Factor and Mounting—The antenna element shall be basically of a horizontal loop design (terminated) and shall be as small as possible commensurate with the performance requirements specified herein, but in no event shall the volume enclosed within the envelope of rotation be larger than $13\frac{5}{8}$ in. in diameter and extend more than $10\frac{21}{32}$ in. from the collar of the TG-50/APA-69 drive unit when the antenna is properly mounted in the drive unit.

3.4.6.3 Contents—The antenna shall contain a directive radiating element and any necessary impedance or transmission line matching device.

3.4.6.4 Weight—The weight of this unit shall not exceed 6.0 lb.

3.4.6.5 Controls—No controls shall be placed on the unit.

3.4.6.6 Electrical Connections—The r-f output from the antenna connects into the CU-174/APA-69 which transfers the r-f energy from the rotating antenna to the antenna feed line to the receiver.

3.4.7 Antenna Assembly AS-676/APA-69—This unit shall meet the following requirements in addition to the applicable requirements of paragraph 3.3:

3.4.7.1 Function—This antenna assembly, when mounted on the TG-50/APA-69 drive unit shall provide directive reception of either horizontally or vertically polarized signals within the frequency range from 180 to 5000 mc.

3.4.7.2 Form Factor and Mounting—The size of this unit shall be such that the volume enclosed within the envelope of rotation shall not exceed 20 in. in diameter and extend more than $11\frac{1}{2}$ in. from the collar of the TG-50/APA-69 drive unit when the antenna is properly mounted in the drive unit.

3.4.7.3 Contents—The unit shall consist of horizontal and vertical unidirectional antennas to cover the frequency range from 180 to 1500 mc and a single reflector with a feed that can be positioned in either a horizontal or a vertical position that will cover the frequency range from 1000 to 5000 mc. Necessary relays and actuators for proper antenna switching shall be included.

3.4.7.4 Weight—The weight of this unit shall not exceed 19.0 lb.

3.4.7.5 Controls—No controls shall be placed on the unit.

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3.4.7.6 Electrical Connections—The r-f output from the antenna connects into the CU-174/APA-69 which transfers the r-f energy from the rotating antenna to the antenna feed line to the receiver.

3.4.8 Antenna Assembly AS-435/APA-69—This unit shall meet the following requirements in addition to the applicable requirements of paragraph 3.3:

3.4.8.1 Function—This unit shall mount on the TG-8A/APA-69A drive unit and shall consist of vertical and horizontal unidirectional antennas covering the frequency range from 140 to 1500 mc. The directivity of the antennas shall be sufficiently good to meet the accuracy of indication requirements specified in paragraph 3.3.3.

3.4.8.2 Form Factor and Mounting—The size of the unit shall not exceed 16.2 in. in diameter by 17.2 in. in depth. It shall mount on the drive shaft of the drive unit by a collar nut which screws onto the drive shaft and clamps the antenna in position.

3.4.8.3 Contents—The unit shall contain the following:

(1) Vertical antenna and reflector

(2) Horizontal antenna and reflector

(3) Antenna polarization switch and actuating relay

(4) Spark suppression capacitor

3.4.8.4 Weight—The weight of this unit shall not exceed 15 lb.

3.4.8.5 Electrical Connections—The r-f output from the antenna is made through the CU-174/APA-69 antenna coupler which connects into antenna receptacle and feeds the r-f through the drive unit.

3.4.8.6 Controls—No controls shall be located on this unit.

3.4.9 Antenna Assembly AS-434/APA-69—This unit shall meet the following requirements in action to the applicable requirements of paragraph 3.3:

3.4.9.1 Function—This assembly shall mount on the TG-8A/APA-69A drive unit and, together with the antenna position information supplied from the drive unit, shall furnish bearing information on either horizontal or vertical polarized received signals in the frequency range from 1000 to 5000 mc. The directivity of the antenna shall be sufficiently good to meet the bearing accuracy indication requirements of paragraph 3.3.3.

3.4.9.2 Form Factor and Mounting—The size of the unit shall not exceed 20 in. in diameter by 18.4 in. in depth. It shall mount on the drive shaft of the drive unit by a collar nut which clamps the unit into position.

3.4.9.3 Contents—The unit shall contain the following:

(1) Vertical dipole antenna and reflector

(2) Horizontal dipole antenna and reflector

(3) Antenna polarization switch and actuating relay

(4) Spark suppression capacitor

3.4.9.4 Weight—The weight of this unit shall not exceed 12 lb.

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3.4.9.5 Electrical Connections—The r-f output from the antenna connects into the CU-174/APA-69 which transfers the r-f energy from the rotating antenna to the antenna feed line to the receiver.

3.4.9.6 Controls—No controls shall be located on this unit.

3.4.10 Antenna Assembly AS-436/APA-69—This unit shall meet the following requirements in addition to the applicable requirements of paragraph 3.3.

3.4.10.1 Function—This antenna assembly shall consist of a broad band horn antenna for reception of horizontally or vertically polarized signals in the frequency range from 4000 to 10750 mc.

3.4.10.2 Form Factor and Mounting—This unit shall mount in the TG-50/APA-69 drive unit in the same position as the CU-174/APA-69 coupler, and shall replace that unit when installed. This unit shall not exceed 2.5 in. in diameter by 13.2 in. in depth.

3.4.10.3 Weight—The weight of this unit shall not exceed 1.5 lb.

3.4.10.4 Controls—No controls shall be placed on the unit.

3.4.10.5 Electrical Connections—A special receptacle designed to mate with a UG-21B/U plug shall be located on the unit for connection of the antenna cable. The transformation of the received energy from the horn to the coaxial cable shall be made within the base of the horn.

3.4.10.6 Circularity of Polarization—The polarization of the AS-436/APA-69 antenna when used in combination with the AT-182/APA-69 reflector, shall be circular within 5 db at 4450 mc, 4 db at 5000 mc, and 10 db at 10500 mc.

3.4.11 Antenna Reflector AT-182/APA-69—This unit shall meet the following requirements:

3.4.11.1 Function—This unit shall be a reflector in the form of a parabolic section in the horizontal plane and straight in the vertical plane. It shall rotate in such an angle as to reflect energy into the AS-436/APA-69 antenna assembly.

3.4.11.2 Form Factor and Mounting—This unit shall mount on the TG-8A/ or the TG-50/APA-69 drive unit in the same manner as the AS-775/APA-69, AS-676/APA-69, AS-435/APA-69, and the AS-434/APA-69 antenna assemblies and shall be rotated by the drive unit. Its size shall not exceed 11.5-in. long, 9.4-in. wide, and 10.9-in. high.

3.4.11.3 Weight—The weight of this unit shall not exceed 2.5 lb.

3.4.11.4 Controls—No controls shall be placed on the unit.

3.4.11.5 Electrical Connections—No electrical connections shall be made to the unit.

3.4.12 Antenna Coupler CU-174/APA-69—The antenna coupler shall furnish r-f energy from rotating antenna assemblies AS-775/APA-69, AS-676/APA-69, AS-435/APA-69, and AS-434/APA-69 through a rotating coaxial joint to the antenna cable.

3.4.12.1 Form Factor and Mounting—This unit shall mount in the center of the drive unit by means of spring clips. The size of the unit shall not exceed 2.3 in. in diameter by 8.3-in. long.

3.4.12.2 Electrical Connections—Electrical connections to the unit shall consist of a coaxial socket which energizes the r-f output receptacle of the antenna when the antenna is mounted to the drive unit, and a special receptacle designed to mate with a UG-21B/U plug for connection of the antenna cable to the receiver.

3.4.12.3 Controls—No controls shall be located on the unit.

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3.4.12.4 Standing Wave Ratio and Insertion Loss—For consideration of standing wave ratio, this unit shall be considered as a part of antenna assemblies AS-775/APA-69, AS-676/APA-69, AS-435/APA-69, and AS-434/APA-69 and the combination shall fulfill the requirements of paragraph 3.3.6. The insertion loss of the unit shall not exceed 1 db.

3.4.12.5 Weight—The weight of this unit shall not exceed 3 lb.

3.4.13 Mounting MT-1005/APA-69A—This unit shall provide a mounting for the IP-81A/APA-69 azimuth panoramic indicator and shall adequately isolate the latter unit from vibration and shock as required.

3.4.13.1 Form Factor—This unit shall be constructed in accordance with the applicable requirements of Specification MIL-C-172 for the mounting size MS 91405-B1C.

3.4.13.2 Weight—The weight of this unit shall not exceed 4 lb.

4. QUALITY ASSURANCE PROVISIONS

4.1 General—Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests—Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests—Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of the contractor and shall be conducted in accordance with the approved test procedure of paragraph 4.6. Data obtained by the contractor in conducting this test shall be submitted to the BUWEPS for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the BUWEPS shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the BUWEPS.

4.2.2 Service Approval Tests—At the completion of the contractor's demonstration tests and when requested by the BUWEPS, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

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4.2.2.1 Accessory Material—In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipments compliance with applicable requirements. The test data shall include data obtained by the contractor in testing to Specification MIL-T-5422.

4.2.3 Scope of Tests—Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval—Approval of the preproduction model shall be by the BUWEPS upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Production Sample Tests—When requested by the BUWEPS, 1 equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests—This equipment may be subjected to any and all tests the BUWEPS deems necessary to assure that the production equipment is equivalent to the previously approved pre-production sample in design, construction, and performance, and that it meets all applicable requirements.

4.3.2 Production Sample Approval—Approval of the production sample shall be by the BUWEPS upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the BUWEPS. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample—On completion of the production sample tests the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests—The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the BUWEPS shall engage the service of a commercial testing laboratory acceptable to the BUWEPS. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

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(3) Special Tests

4.4.1 Individual Tests—Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product—Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test—Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation. The following tests shall be specifically included:

Tests	Applicable Paragraphs
Antenna Assembly AS-775/APA-69	
1. Antenna Gain	3.3.5
2. Stand Wave Ratio	3.3.6
3. Mechanical Stability	3.3.4
Antenna Assembly AS-676/APA-69	
1. Antenna Gain	3.3.5
2. Stand Wave Ratio	3.3.6
3. Mechanical Stability	3.3.4
Antenna Assembly AS-435/APA-69	
1. Antenna Gain	3.3.5
2. Stand Wave Ratio	3.3.6
3. Mechanical Stability	3.3.4
Antenna Assembly AS-434/APA-69	
1. Antenna Gain	3.3.5
2. Stand Wave Ratio	3.3.6
3. Mechanical Stability	3.3.4
Antenna Assembly AS-436/APA-69	
1. Antenna Gain	3.3.5
2. Stand Wave Ratio	3.3.6
3. Mechanical Stability	3.3.4
Antenna Drive Unit TG-8A/ or TG-50/APA-69	
1. Mechanical Accuracy	3.3.2
2. Rotational Speed	3.3.4

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Azimuth-Panoramic Indicator IP-81A/APA-69A	
1. Signal Amplification (DF)	3.4.3.8
2. Bearing Accuracy	3.4.3.9
3. Signal Amplification (Panoramic)	3.4.3.7
4. Frequency Indication	3.4.3.10
Direction Finder Control C-527A/APA-69	
1. Operation of the Controls	3.4.4.4
2. Antenna Speed	3.3.4
Direction Finder Control C-1321/APA-69	
1. Operation of the Controls	3.4.5.4
2. Antenna Speed	3.3.4

4.4.1.3 Manufacturing Reliability Test—Each equipment shall be operated under the conditions specified below for a period of 6 hours without a failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only such adjustments shall be permitted which can be made using controls and adjustments which would be accessible during flight.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any frequency within the range from 25 to 60 cps with a minimum of 0.025 double amplitude

A brief operational check shall be conducted periodically to determine satisfactory operations. The 6 hour testing period may be composed of short periods of 3 hours or more. Should a failure occur, it shall be repaired and the test started over. A record shall be made of all failures.

4.4.2 Sampling Tests—Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests—As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

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4.4.3 Special Tests—Special tests shall be conducted on a quantity of equipments as approved or requested by the BUWEPS for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule—Special tests shall be conducted on the first and every fifth equipment that successfully passes the sampling test. However, additional equipments may be selected when required by the following:

(1) After an engineering change

(2) Whenever failure reports or other information indicates additional tests are necessary

4.4.3.2 Scope of Tests—Special tests shall consist of such tests as are approved by the BUWEPS. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the BUWEPS for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature—Altitude Tests

(2) Vibration Test 60 min cycling in each of 3 places

(3) Humidity Tests

4.4.4 Equipment Failure—Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the BUWEPS for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test—A 300-hr life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following: When a production sample equipment is submitted to the BUWEPS for test, then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

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4.5.1 Test Conditions—The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 to 5000 ft)
Humidity	Room ambient
A-C voltage	115 ± 5 v (at lowest applicable frequency)
D-C voltage	27.0 ± 2.0 v

4.5.2 Test Periods—The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check—At approximately 8-hr intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the BUWEPS.

4.5.4 Test Data—The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) Circuit reference symbol number

(3) Part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) Number of hours which the part operated prior to failure

4.5.4.1 Failure Report—In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the BUWEPS upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The BUWEPS will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples—An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures—The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the contractor and submitted to the BUWEPS for review and approval. The right is reserved by the BUWEPS or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the BUWEPS.

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4.7 Presubmission Testing—No item, part of complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest—Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the BUWEPS.

4.9 Equipment Changes—After approval of the preproduction model, no changes which affect weight, installation interchangeability, and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the BUWEPS. To obtain BUWEPS approval, the contractor shall submit an Engineering Change Proposal giving the necessary information concerning the change. This shall apply for either a BUWEPS' recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change and a Material Bulletin to cover the approved change. Engineering Change Proposals, Material Changes, and Material Bulletins shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General—All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values—Normal and limiting values of performance data shall be determined at input voltages of 27.5 10.5-vdc and 115 11.0-vac. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Precedence of Documents—When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract—The contract shall have precedence over any specification.

(2) This Specification—This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications—Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives—Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the BUWEPS for consideration. Each request shall be accompanied by complete supporting information.

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6.4 For the purpose of this specification the AN/ALR-8 countermeasures receiving set shall be considered to consist of components from the AN/APR-9 radar set or AN/APR-13 receiving equipments.

NOTICE—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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